

Metaheuristic-Based Instance Selection for Compact Training of Deep Neural Networks

Juan Manuel Rodríguez Gómez

*Master's Degree in Artificial Intelligence Research
Menéndez Pelayo International University
Spain*

100013269@alumnos.uimp.es

Supervisor: Pablo Mesejo Santiago

*Department of Computer Science
and Artificial Intelligence
University of Granada
Granada, Spain
pmesejo@ugr.es*

ORCID: 0000-0001-9955-2101

Supervisor: Óscar Cordón García

*Department of Computer Science
and Artificial Intelligence
University of Granada
Granada, Spain
ocordon@decsai.ugr.es*

ORCID: 0000-0001-5112-5629

Abstract—Training image classifiers based on deep learning on large datasets can be computationally demanding, although many examples may be redundant or only marginally informative. This work studies fixed-cardinality instance selection under a linear evaluation protocol with frozen features. Random Search (RS), Large Neighborhood Search (LNS), a Genetic Algorithm (GA), and a Memetic Algorithm (MA) are first compared using cached representations pretrained on ImageNet, with the deep feature extractors kept frozen and only a newly initialized linear classifier trained for each candidate subset. A two-stage experimental design is adopted. The preliminary MNIST stage compares four frozen representations—AlexNet, ResNeXt-50, EfficientNetV2-S, and Swin-T—over five instance retention levels and five seeds with a screening budget of 100 candidate evaluations. AlexNet with GA achieves the highest aggregate validation accuracy (96.976%) and is selected descriptively for the main stage. On CIFAR-10 and Tiny ImageNet, GA and RS are evaluated with an enlarged budget of 1000 candidate evaluations and compared with a Single Stratified Random Sample (SSRS), global k-center greedy (KCG), and the 100% base training baseline. No subset selection method dominates across datasets and retention levels. On CIFAR-10, GA obtains the best mean block rank and reaches 80.87% accuracy at 90% retention, only 0.08 percentage points below the 100% base training result. On Tiny ImageNet, KCG reaches 45.72% accuracy at only 25% retention, 0.23 points below the 100% base training baseline, while SSRS remains competitive at negligible selection cost. Overall, substantial data reduction can preserve competitive performance under linear evaluation, while expensive wrapper search provides only small and inconsistent gains over simpler baselines.

Index Terms—deep learning, image classification, training data reduction, instance selection, transfer learning, metaheuristics, evolutionary computation

I. INTRODUCTION

Deep neural networks (DNNs) have become central to image classification because they learn hierarchical visual representations directly from data [1]. Competitive systems often rely on large labeled datasets and repeated training procedures, increasing computational, storage, data management, and energy requirements. These costs are especially relevant when several configurations and random initializations must be evaluated. Moreover, training examples do not necessarily contribute equally: datasets can contain redundant, noisy, overly simple, or marginally informative observations. Consequently, increasing dataset size does not always produce a proportional improvement in predictive performance. Recent data pruning research has shown that informative subsets can provide more

favorable trade-offs between performance and data retention than equally sized random subsets [2].

Instance selection addresses this problem by retaining a representative subset of the original observations while discarding examples whose contribution is limited [3]. Unlike dataset distillation, which optimizes synthetic examples to summarize a dataset [4], instance selection preserves original images and labels. The resulting subsets remain directly inspectable and can be used by a conventional supervised learning pipeline.

Selecting an effective subset is a difficult combinatorial problem. Given N base training instances and a required subset of size k , there are $\binom{N}{k}$ feasible subsets. Exhaustive evaluation is therefore impractical for realistic image datasets, both because of the size of the search space and the cost of evaluating candidate solutions. Metaheuristics are suitable for this setting because they can explore a discrete search space without requiring a differentiable objective function and can balance diversification with local improvement. Evolutionary and memetic methods have previously been applied successfully to prototype and instance selection [5].

Within this context, this work investigates fixed-cardinality instance selection for image classification based on deep neural representations. The empirical study considers four architectures pretrained on ImageNet—AlexNet, ResNeXt-50, EfficientNetV2-S, and Swin-T—and three datasets of increasing visual complexity: MNIST, CIFAR-10, and Tiny ImageNet. The architectures are used as frozen feature extractors. Thus, the study analyzes how the composition and size of a retained training subset affect a linear classifier trained over a fixed deep representation rather than claiming to reduce the cost of end-to-end DNN optimization.

Four heuristic and metaheuristic search strategies are examined in the preliminary stage. Random Search (RS) independently samples subsets of fixed size and provides an intentionally simple unguided baseline. Large Neighborhood Search (LNS) repeatedly performs randomized destroy-and-repair moves that exchange multiple selected and unselected instances. A Genetic Algorithm (GA) maintains a population and uses tournament selection, crossover that preserves cardinality, swap mutation, and elitism preserving one individual. A Memetic Algorithm (MA) extends GA by probabilistically applying local refinement to offspring. Every method preserves the requested subset cardinality, considers the same fitness

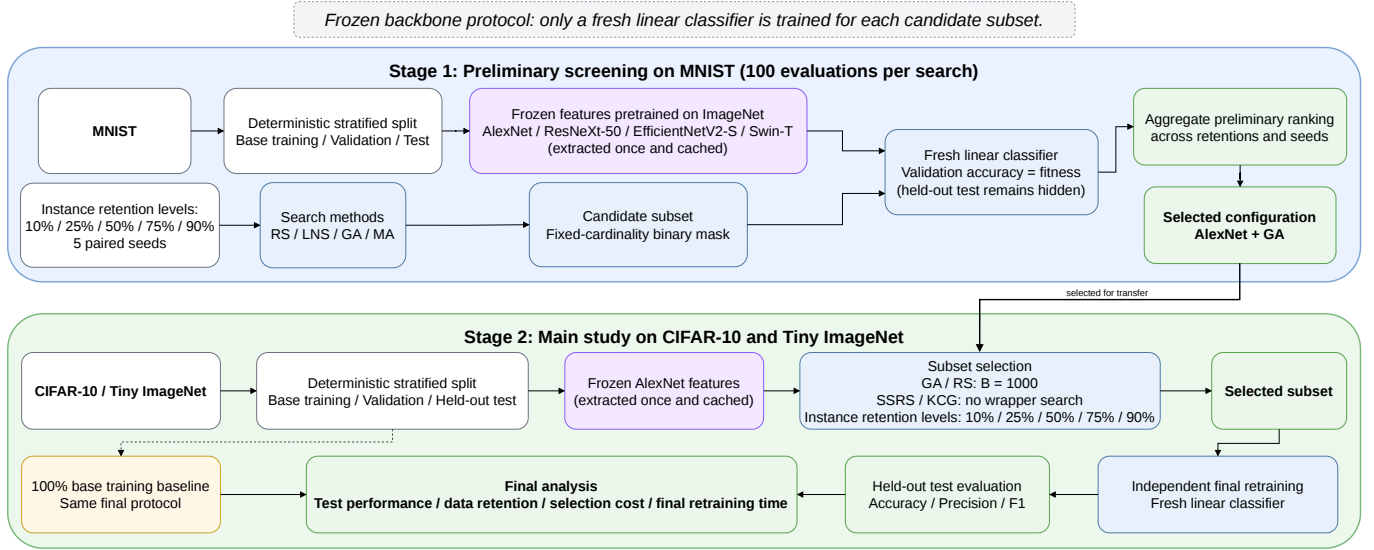


Fig. 1. Overview of the two-stage experimental framework. Stage 1 screens four frozen representations and RS, LNS, GA, and MA on MNIST using five retention levels, five paired seeds, and $B = 100$, selecting AlexNet with GA descriptively. Stage 2 compares GA and RS with $B = 1000$ against SSRS, KCG, and the 100% base training reference on CIFAR-10 and Tiny ImageNet. Selected subsets are independently retrained and evaluated once on held-out test data.

function and receives the same candidate evaluation budget within the preliminary comparison.

The main practical difficulty is fitness cost. Training a complete deep network for each candidate would require a prohibitively large number of expensive optimization processes. This study therefore adopts a controlled transfer learning protocol [6]. A feature extractor pretrained on ImageNet is frozen, its activations are computed once and cached, and each candidate subset is evaluated by training a fresh linear classifier on the selected feature vectors. Frozen representations and linear evaluation are established tools for assessing transferred visual features while greatly reducing the number of trainable parameters [7].

Within each guided search run, every candidate uses the same initial classifier weights, data partition, training seed, optimizer configuration, and epoch budget. Candidate fitness is calculated exclusively on an internal validation partition. The preliminary MNIST searches use $B = 100$ evaluations as a computational screening budget. After this stage identifies the configuration transferred to the main experiments, GA and RS are rerun on CIFAR-10 and Tiny ImageNet with an enlarged budget of $B = 1000$. The preliminary stage does not access held-out test data. In the main stage, once each search has ended and the mask with the highest validation fitness has been fixed, that subset is retrained from an independent classifier initialization and evaluated once on the held-out test partition. The same held-out test partition is used for the final evaluation of every main stage method, retention level, and seed, but it is never used to calculate fitness or to alter the selected mask.

A two-stage incremental experimental design is adopted because a full factorial evaluation over all datasets, feature extractors, algorithms, retention levels, and seeds would be impractical. The preliminary stage uses MNIST to screen the

four representations and four search strategies. The main stage then fixes the selected AlexNet representation and evaluates the selected GA together with three complementary references: RS under the same budget of 1000 evaluations, a Single Stratified Random Sample (SSRS) that does not use validation to guide selection, and global k-center greedy (KCG), an established geometric core set baseline based on farthest-first traversal [8], [9]. The 100% base training set provides the final reference without data reduction. SSRS directly tests whether expensive wrapper search improves over drawing one balanced random subset, whereas KCG tests whether coverage in frozen feature space provides a useful alternative selection principle.

Figure 1 summarizes the complete two-stage framework. The main contributions are:

- A reproducible fixed-cardinality formulation in which candidate subsets are represented by binary masks and optimized through random, large neighborhood, evolutionary, and memetic search.
- A controlled fitness protocol based on cached pretrained features, identical classifier initialization within each run, deterministic execution, and strict separation of selection driven by validation performance from held-out test evaluation.
- A two-stage empirical study comprising 400 preliminary MNIST searches with $B = 100$, followed by 100 definitive GA/RS searches with $B = 1000$, together with SSRS, KCG, and 100% base training baselines on CIFAR-10 and Tiny ImageNet.
- An analysis of predictive performance, instance retention level, statistical robustness, wrapper search convergence, selection overhead, final retraining time, and the practical value of expensive search relative to simpler baselines.

The objective is not to claim that one universally optimal subset or search method exists. Instead, the study asks whether guided search based on validation performance provides a meaningful advantage over unguided sampling and geometric selection, and quantifies the trade-off among retained data, held-out performance, selection cost, and downstream linear classifier training time.

The remainder of this paper is organized as follows. Section II introduces the required background. Section III reviews related work. Section IV describes the fixed-cardinality formulation, fitness protocol, search strategies, and main stage baselines. Section V presents the experimental setup. Section VI reports and discusses the results. Section VII summarizes the conclusions, limitations, and future research directions.

II. BACKGROUND

This section summarizes the instance selection formulation, the role of metaheuristic optimization, and the linear evaluation protocol with frozen features.

A. Instance Selection

Let a labeled training dataset be

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N, \quad (1)$$

where N is the number of available training instances. Instance selection seeks a subset that preserves predictive information while removing redundant, noisy, or weakly informative examples [3]. A candidate is represented by a binary mask

$$\mathbf{z} = (z_1, z_2, \dots, z_N) \in \{0, 1\}^N, \quad (2)$$

with $z_i = 1$ when instance i is retained. Following classical prototype selection formulations [10], the present study imposes the fixed-cardinality constraint

$$\sum_{i=1}^N z_i = k, \quad (3)$$

and the retention ratio is

$$\rho = \frac{k}{N}. \quad (4)$$

Fixing k allows methods to be compared at the same reduction level, so performance differences are attributable to subset composition rather than subset size. The generic optimization problem is

$$\begin{aligned} \mathbf{z}^* = \arg \max_{\mathbf{z}} \quad & J(\mathbf{z}) \\ \text{subject to} \quad & \mathbf{z} \in \{0, 1\}^N, \\ & \sum_{i=1}^N z_i = k, \end{aligned} \quad (5)$$

where $J(\mathbf{z})$ measures subset usefulness. In this work it is estimated through validation performance after training a classifier on the selected instances; Section IV defines the exact protocol.

B. Metaheuristic Optimization

Even with fixed cardinality, the feasible search space contains

$$|\Omega_k| = \binom{N}{k} = \frac{N!}{k!(N-k)!}, \quad (6)$$

which quickly makes exhaustive enumeration infeasible. Candidate quality is also expensive to evaluate because it requires fitting a classifier. Metaheuristics are therefore appropriate approximate optimization strategies for this discrete, non-differentiable setting [11].

The compared paradigms provide different balances between exploration and exploitation. RS samples independent subsets. LNS modifies several decisions of one incumbent through destroy-and-repair moves [12], [13]. GA explores with a population using selection, crossover, and mutation, whereas MA supplements evolutionary search with local refinement, a combination previously used for prototype and instance selection [5]. Their implemented operators are detailed in Section IV.

C. Transfer Learning and Linear Evaluation

Transfer learning reuses representations learned on a source task to support a target task [6]. Let

$$f_{\theta} : \mathcal{X} \rightarrow \mathbb{R}^d \quad (7)$$

denote a neural feature extractor. For image $\mathbf{x}_i$,

$$\mathbf{h}_i = f_{\theta}(\mathbf{x}_i), \quad \mathbf{h}_i \in \mathbb{R}^d. \quad (8)$$

With a frozen backbone, θ is unchanged and only a linear classifier is trained:

$$\mathbf{o}_i = \mathbf{W}\mathbf{h}_i + \mathbf{b}, \quad (9)$$

where $\mathbf{W}$ and $\mathbf{b}$ are trainable. This linear evaluation protocol is widely used to assess transferred visual representations [7]. Because the backbone is fixed, all features can be computed once and cached, so candidate evaluation operates on stored vectors instead of repeatedly executing the complete network. This also keeps the representation identical across candidate subsets.

The interpretation is correspondingly limited: candidate fitness measures the usefulness of a subset for a linear classifier over a particular frozen representation. It does not establish that the same subset would be optimal under end-to-end fine-tuning.

III. RELATED WORK

Training data reduction has been studied from several complementary perspectives. Classical instance selection methods were originally designed to improve the efficiency and robustness of nearest neighbor classifiers. Subsequent evolutionary approaches formulated the task as a combinatorial optimization problem, while more recent deep learning research has introduced gradient-based subset selection, data pruning scores,

core set construction, and dataset distillation techniques. This section reviews these directions and positions the present work with respect to them.

A. Classical and Evolutionary Instance Selection

Early instance selection methods were closely connected to nearest neighbor classification. The Condensed Nearest Neighbor rule proposed by Hart constructs a reduced set of prototypes intended to preserve the classification behavior of the original training set [14]. In contrast, Wilson’s Edited Nearest Neighbor method removes examples whose labels disagree with their local neighborhood, primarily seeking to eliminate noise and improve class boundary definition [15]. These approaches illustrate the traditional distinction between condensation methods, which reduce storage and prediction cost, and editing methods, which emphasize noise removal and classification quality. A broader review of classical instance selection methods is provided by Olvera-López et al. [3].

Selecting an effective subset from a large data collection remains a combinatorial problem. Evolutionary instance selection addresses this difficulty by encoding a candidate subset as a chromosome, commonly a binary vector, and optimizing its composition through selection, crossover, and mutation. Derrac et al. surveyed evolutionary approaches to instance selection and generation, showing that these techniques can jointly account for predictive quality and data reduction [16].

Within this family, García et al. proposed a memetic approach for evolutionary prototype selection that combines evolutionary population search with an *ad hoc* local improvement procedure [5]. The memetic component intensifies the search around promising subsets while the evolutionary population preserves broader exploration. This work is particularly relevant to the present study because it establishes a direct precedent for combining evolutionary optimization and local refinement in instance reduction problems.

Scalability has also received attention in the evolutionary data reduction literature. Triguero et al. introduced MapReduce for Prototype Reduction (MRPR), an approach that distributes prototype reduction across partitions of large datasets [17]. This line of research shows that the cost of selection itself must be considered when reduction methods are transferred to large-scale problems. Nevertheless, most classical evolutionary studies evaluate selected prototypes with nearest neighbor or conventional machine learning classifiers rather than with deep visual representations.

B. Training Data Selection and Core Sets for Deep Learning

The increasing cost of DNN training has renewed interest in constructing compact training subsets. Existing methods differ mainly in the criterion used to estimate the importance or representativeness of each example.

Some approaches define representativeness in terms of the training objective. Coresets for Accelerating Incremental Gradient descent (CRAIG) selects a weighted subset whose aggregate gradient approximates the gradient of the complete

dataset. Its selection problem is expressed through a submodular facility location objective that can be optimized greedily [18]. Gradient Matching (GRAD-MATCH) follows a related principle but directly selects subsets whose gradients match those computed from the complete training data [19]. These techniques are designed to reduce the number of gradient computations required during model training.

A different research direction estimates sample importance from early training dynamics. Paul et al. introduced the Gradient Normed (GraNd) and Error L2-Norm (EL2N) scores, which rank training examples according to quantities derived from gradient norms and prediction errors during the initial stages of learning [20]. Sorscher et al. subsequently studied how the quality of a pruning metric influences neural scaling behavior and showed that informative subsets can provide more favorable trade-offs between performance and data retention than equally sized random subsets [2].

Geometric core set methods instead seek coverage of a representation space. The classical farthest-first traversal of Gonzalez provides a greedy approximation for the metric k -center problem [8]. Sener and Savarese later formulated batch active learning for convolutional neural networks (CNNs) as a core set problem and used a greedy k -center selection procedure in representation space [9]. KCG has consequently become a common geometric reference in deep learning subset selection studies. The DeepCore library, for example, benchmarks it alongside multiple core set methods and emphasizes that simple random selection remains a strong reference whose performance should not be overlooked [21].

Recent work has also examined whether selected subsets remain useful across models. Tukan et al. proposed a provable construction intended to approximate dataset loss across multiple neural network configurations [22]. More generally, subset quality may depend on the representation, optimization state, gradients, or geometric criterion used for selection. The present wrapper framework instead treats validation performance as a black-box objective. SSRS and KCG then provide low-cost random and geometric references for evaluating whether that expensive search is worthwhile.

C. Dataset Distillation and Alternative Reduction Paradigms

Dataset distillation and dataset condensation pursue the same broad goal of reducing the amount of training data, but differ fundamentally from instance selection. Rather than choosing observations from the original dataset, dataset distillation optimizes a small collection of synthetic examples so that training on them approximates training on the full dataset.

Wang et al. introduced dataset distillation by jointly optimizing synthetic inputs and their associated learning process [4]. Zhao et al. later proposed dataset condensation through gradient matching, in which synthetic examples are optimized so that the gradients they produce resemble those obtained from real training data [23]. A comprehensive review by Lei and Tao organizes subsequent techniques according to their optimization objectives, matching strategies, and application scenarios [24].

Distillation can achieve substantially smaller datasets than direct instance selection because synthetic examples are not constrained to correspond to real observations. This flexibility also introduces a more complex optimization process and can reduce the direct interpretability of the resulting data. In contrast, the approach adopted in this work preserves original images and labels, making the selected subsets immediately compatible with conventional supervised learning pipelines.

Active learning is another related but distinct paradigm. Its primary objective is to determine which unlabeled examples should be annotated under a limited labeling budget. Instance selection assumes that all candidate examples are already labeled and determines which of them should be retained for training. Accordingly, the experimental objectives and cost models of the two settings differ.

D. Positioning of the Present Work

The reviewed literature includes neighborhood-based editing and condensation, evolutionary prototype selection, gradient-based subset selection, training dynamics scores, geometric core sets, and synthetic dataset distillation. The present study is positioned at the intersection of evolutionary instance selection and deep visual representation learning.

Unlike classical evolutionary prototype selection studies, candidate subsets are evaluated using neural representations pretrained on ImageNet rather than a nearest neighbor classifier. Unlike score-based pruning and gradient matching methods, the proposed wrapper framework does not assume a fixed analytical importance criterion. Instead, RS, LNS, GA, and MA optimize the validation performance obtained after training a linear classifier on each candidate subset.

All guided search strategies operate under the same fixed-cardinality constraint and fitness evaluation budget within their corresponding stage. This design separates the effect of subset composition from the effect of subset size and supports a controlled comparison of different exploration and exploitation mechanisms. The same frozen representation, classifier initialization, training configuration, and validation partition are used within each paired experimental context.

The main stage broadens the comparison beyond guided search. RS determines whether GA benefits from evolutionary search rather than from evaluating many subsets. SSRS asks whether evaluating many candidates is necessary at all, because it generates one random subset stratified by class without consulting validation data. KCG provides a representation space coverage baseline that also avoids candidate comparison on validation data. Finally, the 100% base training baseline quantifies the predictive and computational trade-off introduced by data reduction. More complex gradient-based methods remain outside the experimental scope of this study.

IV. METHODOLOGY

This section presents the proposed framework for fixed-cardinality instance selection. The methodology combines a binary fixed-cardinality representation, a controlled fitness function evaluated on validation data, four guided or unguided

search strategies used in the preliminary study, two additional main stage baselines that do not perform wrapper search guided by validation performance, and an independent final evaluation of every selected subset on held-out test data.

A. Methodological Overview

For each dataset, a training partition and an external held-out test partition are first defined according to the protocol described in Section V-B. The training partition is then deterministically divided into a base training partition and an internal validation partition. The former defines the instance selection search space, whereas the latter is used exclusively to evaluate candidate subsets. The external test partition remains inaccessible throughout the optimization process.

Before the search process begins, all images are processed by a feature extractor pretrained on ImageNet whose parameters remain frozen. The resulting feature vectors are stored and reused throughout the experiment. Consequently, evaluating a candidate subset only requires training a fresh linear classifier on the selected feature vectors rather than repeatedly executing and optimizing the complete neural network.

For a predefined instance retention ratio, each search algorithm explores different subsets of the same cardinality. Candidate quality is measured by the predictive performance obtained on the validation partition. Once the evaluation budget is exhausted, the best mask found by each search run is stored. The corresponding subset is subsequently retrained using an independent final initialization and evaluated on the held-out test partition. The same final protocol is applied to the entire base training partition, which serves as the 100% base training baseline.

B. Solution Representation and Feasibility

Using the notation introduced in Section II, a candidate subset is represented by the binary mask $\mathbf{z}$ defined in (2). The mask operates exclusively on the N instances belonging to the base training partition.

Given a requested instance retention ratio ρ , the number of selected instances is calculated as

$$k = \max \left(1, \left\lceil N\rho + \frac{1}{2} \right\rceil \right), \quad (10)$$

where half-up rounding is used to convert the requested ratio into an integer cardinality. Search configurations producing $k = N$ are excluded because the 100% base training set is evaluated separately as the 100% base training baseline.

The feasible search space is therefore

$$\Omega_k = \left\{ \mathbf{z} \in \{0, 1\}^N \mid \sum_{i=1}^N z_i = k \right\}. \quad (11)$$

All initialization, crossover, mutation, and neighborhood operators are designed to preserve membership in Ω_k . Consequently, the search algorithms optimize only the composition of the subset and cannot obtain an advantage by retaining more training instances.

A random feasible solution is generated by sampling k distinct indices uniformly without replacement. The selected positions receive a mask value of one, while all remaining positions receive a value of zero.

C. Validation Fitness Evaluation

Let f_θ denote a frozen feature extractor and let

$$\mathbf{h}_i = f_\theta(\mathbf{x}_i) \quad (12)$$

be the precomputed feature vector associated with the i -th training instance. For a candidate mask $\mathbf{z}$, the active training set in feature space is

$$\mathcal{H}(\mathbf{z}) = \{(\mathbf{h}_i, y_i) \mid z_i = 1\}. \quad (13)$$

A linear classifier g_ϕ is trained exclusively on $\mathcal{H}(\mathbf{z})$. Each candidate evaluation uses a fresh model instance and a fresh optimizer. However, the initial classifier parameters are restored from the same previously generated state. Therefore, all candidate instance subsets associated with the same dataset, architecture, and experimental seed start from identical classifier weights.

The classifier is optimized using the cross-entropy loss

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{H}(\mathbf{z})|} \sum_{(\mathbf{h}, y) \in \mathcal{H}(\mathbf{z})} \log p_\phi(y \mid \mathbf{h}). \quad (14)$$

Every candidate receives the same fixed number of training epochs. Early stopping is not used because it would assign different computational budgets to different subsets. Nevertheless, validation loss is computed after each epoch, and the classifier state achieving the lowest validation loss is retained. At the end of training, that state is restored and evaluated on the validation partition.

The fitness of a candidate is consequently defined as

$$F(\mathbf{z}) = M\left(g_{\hat{\phi}(\mathbf{z})}, \mathcal{D}_{\text{valid}}\right), \quad (15)$$

where M denotes the target classification metric and $\hat{\phi}(\mathbf{z})$ represents the classifier parameters from the checkpoint with the lowest validation loss obtained when training on $\mathcal{H}(\mathbf{z})$. Accuracy is used as the primary optimization criterion, although the implementation also supports precision, recall, and F1 score.

Several controls are introduced to reduce variability unrelated to instance selection. The training seed, initial classifier state, data partition seed, data loading seed, number of epochs, optimizer configuration, and feature representation remain fixed when candidates are compared. The random number generator controlling the search algorithm is maintained separately from the randomness involved in classifier training.

D. Random Search

RS serves as the basic optimization baseline. At each evaluation, an independent solution is sampled uniformly from Ω_k , trained, and evaluated using (15). No information obtained from previous candidates is used to generate subsequent solutions.

After B evaluations, where B is the common fitness evaluation budget, the algorithm returns

$$\mathbf{z}_{\text{RS}}^* = \arg \max_{\mathbf{z} \in \{\mathbf{z}^{(1)}, \dots, \mathbf{z}^{(B)}\}} F(\mathbf{z}). \quad (16)$$

RS provides a necessary reference for determining whether the guided search mechanisms exploit useful structure in the instance subset space or merely benefit from evaluating multiple random combinations.

E. Large Neighborhood Search

LNS extends conventional local search by modifying a broader part of the incumbent solution through consecutive destroy and repair operations [12], [13]. The destroy operation partially removes decisions from the current solution, whereas the repair operation reconstructs a complete feasible solution. This strategy allows the search to examine a larger neighborhood than local moves affecting one variable or one swap.

The randomized fixed-cardinality LNS implemented in this work starts from a uniformly generated feasible solution. At each iteration, a destroy operator removes a fixed number q of currently selected instances and a random repair operator replaces them with the same number of currently unselected instances. The resulting solution therefore preserves the required subset cardinality.

Let

$$I_1(\mathbf{z}) = \{i \mid z_i = 1\}, \quad I_0(\mathbf{z}) = \{i \mid z_i = 0\} \quad (17)$$

denote the selected and unselected index sets of the current solution. The destroy operator samples a set

$$D \subseteq I_1(\mathbf{z}), \quad |D| = q, \quad (18)$$

whose instances are removed from the subset. The repair operator then samples a replacement set

$$R \subseteq I_0(\mathbf{z}), \quad |R| = q, \quad (19)$$

and inserts those instances into the solution. The reconstructed candidate $\mathbf{z}_n$ is therefore defined by

$$I_1(\mathbf{z}_n) = (I_1(\mathbf{z}_c) \setminus D) \cup R. \quad (20)$$

Because $|D| = |R| = q$, the operation changes $2q$ binary positions while preserving exactly k selected instances. The corresponding q -exchange neighborhood can be expressed as

$$\mathcal{N}_q(\mathbf{z}) = \left\{ \mathbf{z}' \in \Omega_k \mid \begin{array}{l} |I_1(\mathbf{z}) \setminus I_1(\mathbf{z}')| = q, \\ |I_1(\mathbf{z}') \setminus I_1(\mathbf{z})| = q \end{array} \right\}. \quad (21)$$

The number of feasible solutions in this neighborhood is

$$|\mathcal{N}_q(\mathbf{z})| = \binom{k}{q} \binom{N-k}{q}. \quad (22)$$

Rather than enumerating this potentially very large neighborhood, one reconstructed candidate is sampled randomly at each iteration. For a current solution $\mathbf{z}_c$ and a reconstructed candidate $\mathbf{z}_n$, the acceptance rule is

$$\mathbf{z}_c \leftarrow \mathbf{z}_n \quad \text{if} \quad F(\mathbf{z}_n) \geq F(\mathbf{z}_c). \quad (23)$$

The non-worsening criterion allows the search to move across fitness plateaus. Rejected candidates still consume one unit of the common fitness evaluation budget. The procedure continues until exactly B solutions, including the initial solution, have been evaluated. The highest fitness candidate encountered during the complete trajectory is returned independently of the final incumbent solution.

F. Genetic Algorithm

GA maintains a population of P fixed-cardinality binary masks. The initial population is generated randomly, and every individual is evaluated before the evolutionary process begins.

Parent selection is performed through two tournaments of size $t = 3$. For the first parent, three distinct population positions are sampled uniformly without replacement, and the individual with the highest fitness among them is selected. The second parent is chosen through another tournament with three individuals, with the population position selected as the first parent excluded from the candidate pool before sampling. This ensures that the two selected parents are distinct while preserving a tournament size of three for each parent selection event.

Let S_1 and S_2 denote the selected index sets of two parents. Their common selected instances are

$$C = S_1 \cap S_2, \quad (24)$$

and their symmetric difference is

$$D = S_1 \triangle S_2. \quad (25)$$

The elements of C are inherited by both offspring. A random subset of $k - |C|$ elements is then sampled from D and assigned to the first offspring. The remaining elements of D are assigned to the second offspring. Since both parents contain exactly k selected instances, this operator produces two feasible children with the same cardinality but a different composition.

Each offspring is subsequently mutated with probability p_m . When mutation is activated, the number of exchanges is calculated as

$$s = \max \left(1, \left\lfloor k\mu + \frac{1}{2} \right\rfloor \right), \quad (26)$$

where μ is the mutation fraction. The operator replaces s selected instances with s previously unselected instances, subject to the maximum number of feasible swaps.

Elitism is applied when constructing each new population by preserving one individual. The best individual from the previous generation is copied without modification and without an additional fitness evaluation. The remaining positions are filled with evaluated offspring. The process terminates immediately when the shared budget of B candidate evaluations has been consumed, even when this occurs before completing a full generation.

G. Memetic Algorithm

MA uses the same population initialization, tournament selection, crossover that preserves cardinality, swap mutation, and the same elitism strategy as GA. Its distinguishing component is the probabilistic application of a local refinement procedure to newly generated offspring.

To balance global exploration and local exploitation, local refinement is not systematically applied to every offspring. Instead, this work follows the selective probabilistic application criterion studied by Lozano et al. [25], whereby each offspring is independently considered for local refinement with a fixed probability. After an offspring has been evaluated, local refinement is activated with probability p_{loc} , provided that at least one fitness evaluation remains available. In the present experiments, this probability is set to $p_{loc} = 0.20$. Starting from the offspring, up to L neighboring solutions are generated sequentially. Each neighbor exchanges q_{loc} selected and unselected instances and is accepted when its fitness is not lower than that of the current locally refined solution:

$$\mathbf{z}_r \leftarrow \mathbf{z}_n \quad \text{if} \quad F(\mathbf{z}_n) \geq F(\mathbf{z}_r). \quad (27)$$

The final refined solution, rather than the original offspring, is inserted into the new population. All evaluations performed during local refinement consume units from the same global budget B used by offspring evaluation. MA therefore receives no additional fitness evaluations relative to the other search methods. Its performance reflects how it distributes the common budget between exploration at the population level and local exploitation. Under the $B = 100$ preliminary screening budget, this shared accounting is particularly restrictive for MA because population initialization and local refinement consume the same limited budget. Consequently, the algorithm has little opportunity to alternate evolutionary search and local improvement, and its preliminary performance should be interpreted as screening whose outcome depends on the evaluation budget rather than as evidence of converged behavior.

H. Single Stratified Random Sampling Baseline

The main stage includes an SSRS baseline to test whether evaluating many candidate subsets is beneficial relative to

drawing one balanced random subset. SSRS performs no validation for selection. For class c , with n_c base training instances, the ideal quota is $a_c^* = kn_c/N$. Integer quotas are obtained through Hamilton’s largest remainder rule so that they sum exactly to k , and the required indices are sampled uniformly without replacement within each class using the experimental seed. Thus, SSRS uses labels only to preserve class proportions and has zero fitness evaluations.

I. Global k -Center Greedy Baseline

KCG provides a geometric core set baseline derived from farthest-first traversal [8] and commonly used in deep learning core set selection [9]. Let C_{t-1} be the centers already selected in cached feature space. After a seeded random first center, the next instance is

$$c_t = \arg \max_{i \notin C_{t-1}} \min_{j \in C_{t-1}} \|\mathbf{h}_i - \mathbf{h}_j\|_2^2. \quad (28)$$

Selection is global and label-free. Raw cached AlexNet features are used without normalization and labels are inspected only after selection for diagnostics. One traversal is generated up to 90% retention, so the five subsets for a given dataset and seed are nested prefixes. The full pairwise squared Euclidean distance matrix is stored in float32 as a memory-mapped file and reused, trading repeated computation for $O(N^2)$ storage.

J. Final Subset Evaluation

The test partition is never used to calculate candidate fitness or to guide the search. Once a search run has finished, the mask with the highest validation fitness is stored and the optimization process is considered complete.

For final evaluation, a new classifier initialization is generated using a seed different from the seed used during subset search. The selected instance subset is retrained under the same transfer learning protocol and the classifier checkpoint with the lowest validation loss is evaluated once on the held-out test partition. This procedure yields test accuracy, precision, recall, F1 score, and loss. In the main stage, this final test evaluation is performed for every method, retention level, and seed, yielding 200 selected subset test evaluations, together with ten evaluations of the 100% base training baseline. The held-out test partition is used only for this final reporting and comparison and never participates in candidate generation, fitness evaluation, or subset selection.

The 100% base training baseline is independently evaluated using the same final initialization, optimizer, number of epochs, validation checkpointing rule, and test partition. This ensures that differences between a selected subset and the 100% base training baseline cannot be attributed to different classifier initializations or training configurations.

The implementation also records the complete candidate evaluation history, the evolution of the best fitness, the best binary mask, execution times, random seeds, software versions, and the complete methodological configuration. A configuration fingerprint is stored with every experiment and is checked when existing artifacts are resumed, preventing results

generated under incompatible methodological settings from being combined.

V. EXPERIMENTAL SETUP

This section describes the research questions, datasets, frozen feature extractors, two-stage experimental design, parameter settings, baselines, evaluation criteria, statistical analysis, and reproducibility measures used in the empirical study.

A. Research Questions

The experimental study addresses the following research questions (RQs):

- RQ1** Under the same screening fitness evaluation budget, how do RS, LNS, GA, and MA compare on MNIST across the four frozen representations, and which model and algorithm configuration provides the strongest aggregate preliminary validation performance for transfer to the main stage?
- RQ2** How does the instance retention level affect validation performance, held-out test performance, and final retraining and evaluation time?
- RQ3** On CIFAR-10 and Tiny ImageNet, does the model and algorithm configuration selected in the preliminary stage provide a consistent advantage over RS, SSRS, or KCG?
- RQ4** What trade-offs in predictive performance, selection cost, and downstream training do compact subsets provide with respect to the 100% base training baseline?

B. Datasets and Data Partitions

The study considers three well-known image classification datasets with increasing numbers of classes and visual complexity: MNIST, CIFAR-10, and Tiny ImageNet. MNIST contains grayscale images of handwritten digits belonging to ten classes [26]. CIFAR-10 comprises color images distributed across ten object categories [27]. Tiny ImageNet contains 200 classes and images with a native resolution of 64×64 pixels [28].

For each dataset, the available data are first organized into a training partition and a held-out test partition. For MNIST and CIFAR-10, the official training and test splits are used directly. Since the official Tiny ImageNet test labels are not publicly available, its original training split is used as the training partition, whereas its labeled official validation split is reserved as the held-out test partition.

The resulting training partition is subsequently divided into a base training partition and an internal validation partition. This split is deterministic, is performed independently within each class, and assigns 20% of the training examples to validation. Half-up rounding determines the number of validation instances per class while ensuring that at least one example from every class remains in both partitions. The base training partition defines the search space of the instance selection methods, whereas the internal validation partition is used exclusively to evaluate candidate subsets. Table I

TABLE I
DATASET PARTITIONS AFTER THE DETERMINISTIC STRATIFIED 80/20 TRAINING/VALIDATION SPLIT. THE OFFICIAL MNIST AND CIFAR-10 TEST SETS ARE HELD OUT; THE LABELED TINY IMAGENet VALIDATION SPLIT SERVES AS ITS HELD-OUT TEST SET.

Dataset	Classes	Base train	Valid.	Test
MNIST	10	48,000	12,000	10,000
CIFAR-10	10	40,000	10,000	10,000
Tiny ImageNet	200	80,000	20,000	10,000

TABLE II
FROZEN FEATURE EXTRACTORS PRETRAINED ON IMAGENet. INPUT SIZE FOLLOWS THE CORRESPONDING TORCHVISION WEIGHTS; FEATURE DIMENSION IS THE OUTPUT AFTER REPLACING THE ORIGINAL CLASSIFIER WITH THE IDENTITY FUNCTION.

Architecture	Input size	Feature dim.
AlexNet	224×224	4096
ResNeXt-50 $32 \times 4d$	224×224	2048
EfficientNetV2-S	384×384	1280
Swin-T	224×224	768

summarizes the resulting base training, validation, and held-out test partitions.

The partitioning seed is fixed to 42. Consequently, all methods operating on the same dataset use exactly the same base training, validation, and test instances. Selection masks refer only to positions within the base training partition and are never applied to validation or test data.

C. Neural Architectures and Feature Extraction

Four architectures pretrained on ImageNet are considered as frozen feature extractors: AlexNet [29], ResNeXt-50 $32 \times 4d$ [30], EfficientNetV2-S [31], and Swin-T [32]. They represent different convolutional and transformer-based design families.

For each extracted representation, the original final classification layer is replaced by the identity function. The stored feature vector is therefore the exact input that would be received by a newly initialized replacement linear classifier. Table II summarizes the input resolution associated with the default pretrained weights and the resulting feature dimension.

The official deterministic preprocessing transformation associated with each Torchvision weight configuration is used. MNIST images are converted from one to three channels before applying the corresponding transformation. No stochastic data augmentation is used during feature extraction.

Feature representations from all four architectures are computed for MNIST to support the complete model and algorithm comparison performed in the preliminary stage. For the main stage, only the representation selected according to the aggregate preliminary validation results is computed for CIFAR-10 and Tiny ImageNet. Feature tensors preserve the original sample order and are stored together with their labels. The deterministic stratified split is subsequently recreated from those labels, ensuring a stable positional mask space.

During candidate evaluation, only a fresh linear classifier is trained. The pretrained extractor is neither executed again

nor modified. Frozen Dropout and BatchNorm modules remain in evaluation mode during feature extraction, and the resulting cached representation is identical for all candidates in the same dataset and architecture context.

D. Two-Stage Experimental Setup

A complete factorial evaluation of all datasets, frozen feature extractors, search algorithms, instance retention levels, baselines, and seeds would produce an impractically large experimental matrix. An incremental two-stage design is therefore used. The first stage is a broad screening experiment with a deliberately low budget on MNIST. The second stage fixes the representation selected in the preliminary study, uses more complex natural image datasets, increases the budget for guided search by one order of magnitude, and adds complementary baselines without wrapper search.

1) *Preliminary Stage*: The preliminary stage is conducted on MNIST and compares the four frozen feature extractors combined with RS, LNS, GA, and MA, producing sixteen model and algorithm combinations. Five fixed instance retention levels are considered:

$$\rho \in \{0.10, 0.25, 0.50, 0.75, 0.90\}. \quad (29)$$

Each model, algorithm, and retention configuration is repeated using five paired search seeds:

$$\mathcal{R} = \{42, 123, 456, 789, 1024\}. \quad (30)$$

Every preliminary search receives exactly $B = 100$ fitness evaluations. Consequently, this screening stage comprises

$$4 \times 4 \times 5 \times 5 = 400 \quad (31)$$

complete search runs, corresponding to 40,000 candidate linear classifier evaluations. Validation accuracy is the optimization objective, and the test partition is not accessed in this stage. The model and algorithm configuration for the main stage is selected using aggregate best validation accuracy across retention levels and seeds, with mean rank across seeds, seed wins, and search time used as complementary descriptive criteria. The instance retention level itself is not selected as a hyperparameter and remains a controlled factor in the main experiments.

The budget of 100 evaluations is intentionally treated as a screening budget rather than as evidence of search convergence. This distinction is important because the combinatorial search spaces are extremely large and because methods based on populations and local refinement can consume a substantial part of a small budget during initialization and early search. The limitation is especially relevant to MA, whose local refinement evaluations are charged to the same global budget as offspring evaluations. The preliminary ranking may therefore favor methods that exploit very small evaluation budgets more efficiently.

2) *Main Stage*: The frozen representation and search method selected from the preliminary stage are transferred to CIFAR-10 and Tiny ImageNet. For each dataset, retention level, and seed, the selected guided method is compared with RS under the same enlarged budget of $B = 1000$ validation fitness evaluations. This produces

$$2 \times 2 \times 5 \times 5 = 100 \quad (32)$$

guided searches and a total of 100,000 candidate evaluations.

Two additional subset baselines are evaluated without fitness search driven by validation performance. SSRS generates one random subset stratified by class for every dataset, retention level, and seed configuration. KCG generates one seeded farthest-first traversal per dataset and seed, from which the five retention levels are obtained as nested prefixes. Each baseline therefore contributes 50 selected subsets:

$$2 \times 5 \times 5 = 50. \quad (33)$$

In total, the main stage contains 200 selected subset final evaluations (100 from the selected guided method and RS, 50 from SSRS, and 50 from KCG), plus ten 100% base training evaluations, one for each dataset and final seed. Every selected subset is evaluated with the independent final seed associated with its search or sampling seed. Test data are accessed only after the subset has been completely determined.

E. Parameter Configuration

Table III reports the common training and selection parameters. The values are treated as fixed experimental design choices rather than as individually optimized hyperparameters. They were chosen to support a controlled comparison under the available computational budget: training settings are kept identical across stages and methods, whereas settings specific to each search are shared whenever the algorithms use the same operators. The principal experimental difference is the candidate evaluation budget: $B = 100$ in the preliminary screening stage and $B = 1000$ for the selected guided method and RS in the definitive main stage. SSRS and KCG do not use validation fitness evaluations.

The values in Table III arise from three considerations. First, the retention levels, paired seeds, validation split, training epochs, batch size, optimizer, learning rate, and weight decay are fixed protocol choices intended to expose the effect of subset composition while keeping classifier training comparable across methods. The five retention levels span aggressive to mild data reduction, and five paired seeds provide repeated observations while keeping the complete experimental matrix computationally manageable. These settings should therefore be interpreted as controlled experimental conditions rather than as claimed optimal hyperparameters.

Second, the evolutionary and local search parameters were selected as compact settings compatible with the limited fitness evaluation budgets. GA and MA share population size $P = 10$, tournament size $t = 3$, mutation probability $p_m = 0.10$, and

TABLE III
EXPERIMENTAL PARAMETER CONFIGURATION. THE $B = 100$ PRELIMINARY BUDGET IS USED ONLY FOR MNIST SCREENING, WHEREAS THE SELECTED GUIDED METHOD AND RS USE $B = 1000$ ON CIFAR-10 AND TINY IMAGENET.

Parameter	Value
Optimization metric	Validation accuracy
Precision/recall/F1 averaging	Macro
Retention levels	10%, 25%, 50%, 75%, 90%
Search repetitions	5 paired seeds
Preliminary fitness budget	100
Main wrapper search fitness budget	1000
SSRS/KCG fitness evaluations	0
Epochs per training run	10
Batch size	32
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	0
Validation fraction	0.20
Population size	10
Tournament size	3
Mutation probability	0.10
Mutation fraction	0.05
LNS destroy/repair size q	10
Memetic refinement probability	0.20
Memetic refinement evaluations	10
Memetic refinement swap pairs	5
KCG distance metric	Squared Euclidean
KCG first center	Seeded random global instance
Final retrainings	1 per subset and seed

mutation fraction $\mu = 0.05$ so that their common evolutionary component is directly comparable. LNS uses $q = 10$ exchanged pairs to define a genuinely large neighborhood while preserving exact subset cardinality. The MA follows the selective probabilistic local refinement scheme discussed by Lozano et al. [25]; the numerical settings $p_{loc} = 0.20$, $L = 10$, and $q_{loc} = 5$ limit the amount of the shared global budget that can be devoted to local refinement. As discussed in Section V-D, the $B = 100$ preliminary budget is a computational screening budget rather than a convergence budget, whereas $B = 1000$ increases the search effort by one order of magnitude in the main stage while remaining computationally feasible.

Third, the KCG configuration follows the standard farthest-first k -center formulation [8], [9]. Squared Euclidean distance is used on the cached frozen features, and a seeded random first center provides a reproducible starting point for the repeated runs. Overall, the parameter configuration prioritizes controlled comparability and computational feasibility over exhaustive parameter tuning.

The Adam optimizer is used with cross-entropy loss [33]. Every candidate and final model is trained for exactly ten epochs. Validation loss is computed after every epoch and the model state with the lowest validation loss is restored before calculating the requested validation or test metrics. This checkpointing rule is not early stopping because every training run completes all ten epochs.

F. Evaluation Criteria

Preliminary search quality is summarized through the best validation accuracy obtained within the screening budget of 100 evaluations, aggregate mean accuracy, mean rank across seeds, seed wins, and search time. Main stage predictive performance is measured on held-out test data using accuracy, macro precision, macro recall, and macro F1 score. Macro recall is computed but omitted from the main result tables because the CIFAR-10 and Tiny ImageNet held-out test sets contain the same number of examples per class. In this balanced setting, macro recall is numerically identical to accuracy.

For a selected subset S , its accuracy difference with respect to the 100% base training baseline is

$$\Delta_{\text{acc}} = \text{Acc}(S) - \text{Acc}(\mathcal{D}_{\text{base}}), \quad (34)$$

where positive values indicate an improvement over the 100% base training result. The data reduction ratio is

$$R_{\text{data}} = 1 - \frac{|S|}{|\mathcal{D}_{\text{base}}|}. \quad (35)$$

The recorded final duration includes DataLoader construction, model and optimizer creation, ten epoch retraining with validation checkpointing, and held-out test evaluation. Its reduction relative to the 100% base training baseline is

$$R_{\text{time}} = 1 - \frac{T_{\text{final}}(S)}{T_{\text{final}}(\mathcal{D}_{\text{base}})}. \quad (36)$$

Selection cost and final pipeline time are reported separately. For the selected guided method and RS, selection cost is the complete $B = 1000$ search time guided by validation performance. For SSRS, it is the time required to construct one stratified mask. For KCG, the one-time shared pairwise distance cache and the seeded farthest-first traversal are reported separately from the inexpensive materialization of retention level prefixes. The one-time extraction of frozen neural features is excluded from these timing comparisons.

For each configuration, the mean and sample standard deviation across five seeds are reported. Accuracy differences and time reductions are calculated after pairing each selected subset result with the 100% base training baseline that uses the same final seed.

G. Statistical Analysis

The search or sampling seed is treated as the independent replication unit. Only five paired seeds are available, which imposes a severe resolution limit on pairwise nonparametric tests. For a two-sided exact Wilcoxon signed-rank test with five nonzero paired differences, the smallest attainable p -value is

$$p_{\min} = \frac{2}{2^5} = 0.0625. \quad (37)$$

Consequently, pairwise significance at the conventional $\alpha = 0.05$ level is mathematically unattainable. The tests are retained as transparent diagnostics, while interpretation focuses

primarily on mean differences, sample standard deviations, ranks, wins, and the consistency of behavior by retention level.

In the preliminary stage, best validation accuracy is averaged across the five retention levels for each method and seed. A Friedman test compares the sixteen method and representation combinations using the aggregates computed for the five seeds. Two-sided Wilcoxon comparisons between the selected configuration and the fifteen competitors are adjusted using Holm's procedure. The smallest possible first Holm-adjusted value is therefore

$$15 \times 0.0625 = 0.9375, \quad (38)$$

so no preliminary pairwise comparison can attain significance under the present design.

In the main stage, held-out test accuracy is first averaged across the five retention levels for the selected guided method, RS, SSRS, and KCG within each seed. A Friedman test is used as an omnibus diagnostic for the four paired methods within each dataset. The six pairwise Wilcoxon comparisons are then Holm-corrected separately for CIFAR-10 and Tiny ImageNet. With five seeds and six comparisons, the smallest possible first adjusted value is

$$6 \times 0.0625 = 0.375. \quad (39)$$

Thus, even when the omnibus ranking differs across methods, the present pairwise analysis cannot localize a difference at $\alpha = 0.05$. Means by retention level and win/loss counts over the 25 retention level and seed blocks are consequently treated as descriptive evidence.

H. Reproducibility and Computational Environment

The training/validation split seed is fixed to 42. For a search seed r , the candidate training seed is

$$r_{\text{training}} = r + 100000, \quad (40)$$

and the independent final evaluation seed is

$$r_{\text{final}} = r + 200000. \quad (41)$$

Within one guided search run, every candidate starts from the same stored classifier initialization and uses the same training data shuffle seed. The pseudorandom generator controlling the selection algorithm is independent of the generator used for classifier initialization and training.

Python, NumPy, and PyTorch random number generators are seeded. Deterministic PyTorch algorithms are enforced, cuDNN benchmarking is disabled, deterministic cuDNN behavior is enabled, and CUBLAS_WORKSPACE_CONFIG is set to :4096:8. Experimental DataLoaders use zero worker processes, while feature extraction uses four.

The experiments were executed as independent Slurm jobs on the high performance computing infrastructure of the Andalusian Research Institute in Data Science and Computational Intelligence (DaSCI). The preliminary MNIST search stage ran on Hera (two Intel Xeon 4114 processors, 128 GB DDR4,

TABLE IV

AGGREGATED MNIST PRELIMINARY RESULTS. ACCURACY AND SEARCH TIME ARE AVERAGED OVER 25 RUNS PER COMBINATION; RANK AND WINS ARE COMPUTED AFTER AVERAGING RETENTION LEVELS WITHIN EACH SEED. BOLDFACE MARKS THE BEST VALUE IN EACH COLUMN.

Model	Alg.	Acc. (%)	Rank	Seed wins	Time (min)
AlexNet	GA	96.976	1.60	3	37.77
AlexNet	RS	96.944	2.70	0	37.68
AlexNet	MA	96.944	3.10	0	36.27
ResNeXt-50	GA	96.918	3.40	2	34.65
AlexNet	LNS	96.891	4.80	0	37.59
ResNeXt-50	MA	96.882	5.60	0	35.88
ResNeXt-50	RS	96.843	6.80	0	36.30
ResNeXt-50	LNS	96.768	8.00	0	35.76
Swin-T	MA	96.602	10.30	0	26.01
Swin-T	GA	96.598	10.30	0	26.54
EfficientNetV2-S	GA	96.596	10.40	0	35.42
EfficientNetV2-S	MA	96.567	12.20	0	35.61
Swin-T	RS	96.562	12.40	0	27.64
EfficientNetV2-S	RS	96.548	13.80	0	35.62
Swin-T	LNS	96.507	14.80	0	26.09
EfficientNetV2-S	LNS	96.485	15.80	0	35.46

four NVIDIA Titan RTX graphics processing units (GPUs)), whereas all Stage 2 selection and final evaluation jobs ran on Zeus (two Intel Xeon E5-2630 processors, 128 GB DDR3, four NVIDIA GeForce RTX 2080 Ti GPUs). Each selection or final evaluation job used one GPU and four central processing unit (CPU) cores. Requested system memory was 20 GB for the preliminary stage, 36 GB for main wrapper search jobs, 32 GB for KCG, 8 GB for SSRS, and 20 GB for the 100% base training baseline. Keeping all definitive Stage 2 selection and final evaluation jobs on Zeus makes the reported timing measurements internally comparable. The stored manifests report Python 3.12.13, PyTorch 2.5.1+cu121, Torchvision 0.20.1+cu121, NumPy 2.5.1, and scikit-learn 1.9.0.

Each run stores its complete configuration, random seeds, selected mask, candidate history, best fitness trajectory, execution times, and software versions. A SHA-256 configuration fingerprint is checked when experiments are resumed so that incompatible artifacts are not combined. The implementation, experiment scripts, and result processing code are available at <https://github.com/jmarg6/TFM>.

VI. RESULTS AND DISCUSSION

This section presents and discusses the preliminary and main experimental results according to the research questions defined in Section V-A.

A. Preliminary Experimental Results

1) *Validation Performance:* Table IV summarizes the sixteen preliminary model and algorithm combinations over five retention levels and five paired seeds.

AlexNet produced the strongest overall results: its four algorithmic variants occupied four of the five highest positions by mean validation accuracy. AlexNet with GA obtained the highest aggregate accuracy, 96.976%, and the best mean rank across seeds, 1.60. It was the configuration with the best rank for three seeds, while ResNeXt-50 with GA was highest for the remaining two.

The differences among the leading combinations were small. AlexNet with RS and MA both achieved 96.944%, only 0.032 percentage points below AlexNet with GA. Consequently, the preliminary ranking identifies a leading group rather than a large separation between the first positions. Given the deliberately limited $B = 100$ screening budget, this small margin should not be interpreted as robust evidence that AlexNet with GA is intrinsically the best configuration. Instead, it is treated as a pragmatic screening choice for transfer to the more computationally expensive main stage.

At each retention level, GA obtains the highest mean validation accuracy at the three most restrictive conditions: AlexNet + GA reaches 95.615% at 10% and 96.722% at 25%, while ResNeXt-50 + GA reaches 97.307% at 50%. AlexNet + MA is best at 75% (97.600%), and AlexNet + RS is best at 90% (97.707%).

Within AlexNet, all four methods improve as the retained data percentage increases and the effect of retention is substantially larger than the differences among search strategies. GA is best at 10%, 25%, and 50%, whereas MA and RS obtain slightly higher means at 75% and 90%, respectively.

2) *Statistical and Computational Analysis:* After averaging the five instance retention levels within each seed, the Friedman test detected differences among the sixteen combinations:

$$\chi_F^2(15) = 71.01, \quad p < 0.001. \quad (42)$$

The fifteen two-sided Wilcoxon comparisons between AlexNet with GA and the remaining configurations produced a minimum Holm-adjusted p -value of 0.938. This value is effectively the theoretical lower bound of 0.9375 imposed by five paired seeds and fifteen Holm-corrected comparisons. Therefore, the absence of significant *post-hoc* results is not an informative experimental finding: statistical significance was impossible under this design, even if one configuration had won in all five seeds. The *post-hoc* values are retained only for transparency and must not be interpreted as evidence that the methods are equivalent. The preliminary selection is instead based on descriptive aggregate accuracy, mean rank, seed wins, observed margins, and search time.

Mean search time ranged from approximately 26 to 38 minutes. AlexNet with GA required 37.77 minutes on average, which was comparable to the other AlexNet searches. Its higher aggregate validation accuracy was therefore not associated with a distinct search time penalty within the preliminary AlexNet comparison.

3) *Selection of the Model and Algorithm Configuration:* AlexNet with GA was selected for the main experimental stage because it obtained the highest aggregate validation accuracy, the best mean rank across seeds, and three of the five seed wins under the predefined screening protocol. The difference from AlexNet with RS and MA was nevertheless only 0.032 percentage points. Because the *post-hoc* tests were structurally unable to attain $p < 0.05$, they cannot confirm or refute pairwise superiority. The choice of AlexNet with GA should therefore be understood as a descriptive and

computationally pragmatic screening decision used to reduce the experimental matrix before the more expensive CIFAR-10 and Tiny ImageNet stage, rather than as evidence that this configuration is intrinsically or universally superior. AlexNet was consequently fixed as the frozen feature extractor and GA as the selected search method for the main stage. The five instance retention ratios remain controlled experimental conditions in the main experiments.

B. Main Experimental Results

The definitive main stage uses the frozen AlexNet representation and compares GA, RS, SSRS, and KCG on CIFAR-10 and Tiny ImageNet. GA and RS each receive 1000 validation fitness evaluations per run, whereas SSRS and KCG construct subsets without wrapper search guided by validation performance. Five retention levels and five seeds yield 200 test evaluations of selected subsets, together with ten 100% base training results.

1) *Held-Out Test Performance*: Figure 2 summarizes held-out test accuracy for all four subset selection methods and the 100% base training baseline. Macro precision and macro F1 score were also computed. They lead to the same broad interpretation and are retained in the released result tables.

On CIFAR-10, GA obtains the highest mean subset accuracy at 10%, 25%, and 90% retention, reaching 80.87% at 90%. RS is best at 50% with 80.21%, while SSRS is best at 75% with 80.42%. KCG is competitive at the larger retentions but does not produce the highest mean at any CIFAR-10 retention level. The 100% base training baseline reaches 80.95%.

Tiny ImageNet presents a more heterogeneous pattern. SSRS is best at 10%, KCG at 25%, RS at 50% and 90%, and GA at 75%. The most striking result is KCG at 25% retention: 45.72% accuracy, only 0.23 percentage points below the 45.95% result of the 100% base training baseline. However, the same KCG traversal is strongly non-monotonic in predictive performance: its mean accuracy falls to 42.72% at 50% before increasing again at higher retentions. Because the masks are nested prefixes and the geometric covering radius decreases as centers are added, this behavior shows that improved coverage under the k-center objective does not imply monotonic downstream linear classifier accuracy. It also illustrates that the geometry optimized by KCG is not perfectly aligned with the validation/test objective used by the wrapper methods.

2) *Descriptive Ranking and Statistical Diagnostics*: The main stage rankings confirm that the method ordering depends on the dataset. On CIFAR-10, GA obtains the best mean rank over the 25 retention level and seed blocks (1.80) and 11 block wins, followed by RS (2.52, eight wins). KCG and SSRS both have mean rank 2.84. On Tiny ImageNet, RS has the best mean rank (2.40), closely followed by SSRS (2.44), GA (2.52), and KCG (2.64). KCG nevertheless records eight block wins on Tiny ImageNet because its large advantage at 25% retention is concentrated in a subset of the experimental conditions.

The Friedman diagnostics based on seed aggregates are

$$\chi_F^2(3) = 9.00, \quad p = 0.0293 \quad (43)$$

for CIFAR-10 and

$$\chi_F^2(3) = 9.72, \quad p = 0.0211 \quad (44)$$

for Tiny ImageNet. These omnibus results indicate differences in the method rankings, but they must be interpreted together with the severe resolution limit imposed by five seeds. None of the six pairwise Wilcoxon comparisons within either dataset can reach $p < 0.05$ before correction, and no Holm-adjusted comparison is significant. The smallest adjusted value is 0.375, which is exactly the theoretical lower bound induced by six comparisons and five paired seeds.

Descriptively, GA exceeds RS by 0.188 percentage points on average over the CIFAR-10 blocks and wins 16 of 25 comparisons. Its mean advantages over SSRS and KCG are 0.338 and 0.382 points, respectively, with 19–6 and 20–5 win/loss counts. On Tiny ImageNet, the aggregate differences are smaller and less stable. GA exceeds RS by only 0.151 points and wins 12 of 25 blocks, while SSRS and RS differ by only 0.068 points on average. KCG has a 0.359-point aggregate advantage over GA despite losing 13 of the 25 direct blocks. This apparent contradiction is explained by the unusually large KCG gain at 25% retention. These results reinforce the need to inspect behavior by retention level rather than relying on one global average.

3) *Retention, 100% Base Training Trade-Offs, and Final Training Time*: Increasing retention generally improves predictive performance, but the additional baselines show that the relationship is not strictly monotonic for every selection rule. More importantly, several compact configurations approach the 100% base training result closely.

On CIFAR-10, GA at 90% retention is only 0.082 percentage points below the 100% base training baseline and reduces final retraining and evaluation time by 8.0%. A more aggressive compromise is RS at 50%, which reaches 80.21%, 0.74 points below the 100% base training baseline, while reducing the final pipeline time by 43.5%. SSRS at 75% reaches 80.42%, only 0.53 points below the 100% base training baseline, with an 18.6% mean time reduction.

On Tiny ImageNet, KCG at 25% provides a particularly favorable observed trade-off between performance and data retention: 45.72% accuracy, only 0.23 points below the 100% base training baseline, together with a 60.9% reduction in final retraining and evaluation time. At 50%, RS reaches 43.67%, a 2.28-point loss, with a 40.9% final time reduction. At high retention, the measured time differences become small relative to execution noise and can occasionally be slightly negative. Therefore, timing values at high retention should not be interpreted as speed effects specific to a method.

The practical benefit of compact data is consequently clear at the final linear classifier stage, but the magnitude of that benefit depends on the acceptable performance loss and on whether selection cost is amortized over future reuse of the selected subset.

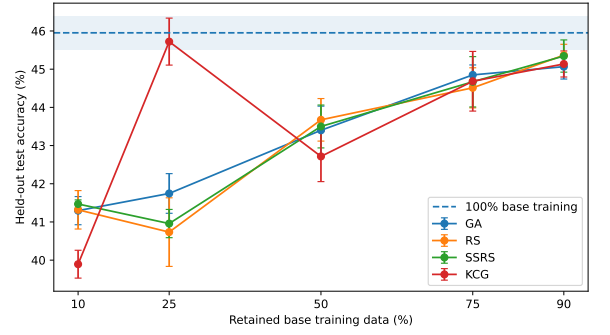
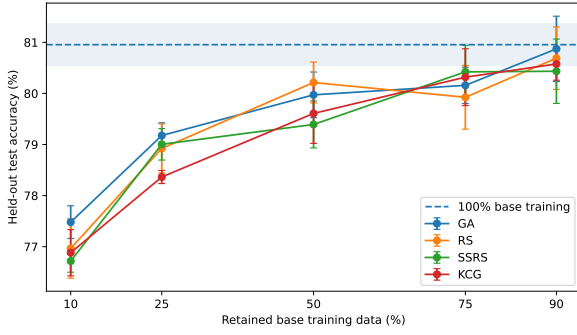


Fig. 2. Held-out test accuracy as a function of the retained base training percentage for GA, RS, SSRS, and KCG using frozen AlexNet features. The dashed horizontal line denotes the mean 100% base training result and the shaded band represents one sample standard deviation across the five paired seeds. The left panel shows CIFAR-10 and the right panel Tiny ImageNet. No subset selection method dominates at every retention level.

4) Selection Cost and Effect of the Enlarged Search Budget:

The definitive GA/RS searches are substantially more expensive than the final retraining they are intended to accelerate. Mean $B = 1000$ search times range from approximately 99 to 448 minutes on CIFAR-10 and from 194 to 844 minutes on Tiny ImageNet, increasing with the retained subset size. Across all subset methods and the 100% base training reference, mean final retraining and test durations range from approximately 6–28 seconds on CIFAR-10 and 11–54 seconds on Tiny ImageNet.

SSRS has negligible selection cost, with mean mask construction below 0.06 seconds. KCG is also much cheaper than wrapper search after a shared geometric preprocessing step: its full pairwise distance cache takes 295.5 seconds on CIFAR-10 and 1156.8 seconds on Tiny ImageNet, while the first seeded traversal to 90% takes 94.2 and 337.9 seconds, respectively. These reusable costs are reported separately from the retention level prefixes.

The enlarged budget materially changes the search outcome. Across the 100 definitive GA/RS runs, only 10% had found their eventual $B = 1000$ best validation fitness by evaluation 100, compared with 54% by evaluation 500 and 90% by evaluation 750. The preliminary $B = 100$ setting should therefore be interpreted as a screening budget, not as a convergence budget.

VII. CONCLUSIONS AND FUTURE WORK

This work studied fixed-cardinality instance selection using frozen representations pretrained on ImageNet. The preliminary MNIST experiments answer **RQ1**: under the common $B = 100$ screening budget, the leading model and algorithm combinations achieved similar validation performance, although AlexNet with GA obtained the highest aggregate accuracy (96.976%), the best mean rank across seeds, and three of the five seed wins. AlexNet with GA was therefore selected for the main stage based on descriptive aggregate performance. Its margin over AlexNet with RS and MA was only 0.032 percentage points, and the design with five seeds prevented pairwise statistical superiority from being established. The very small separation should also be interpreted in

light of the deliberately restricted $B = 100$ screening budget, which may have limited the opportunity for the search methods to differentiate their behavior. The selected configuration is therefore a computationally pragmatic screening choice whose outcome depends on the evaluation budget rather than evidence of intrinsic algorithmic superiority.

The experiments also address **RQ2** by showing how instance retention affects predictive performance and final retraining cost. Increasing the retained data percentage generally improved predictive performance, although the relationship was not strictly monotonic for every method and dataset. More aggressive data reduction could substantially reduce final retraining and evaluation time, while the benefit decreased as retention approached 100% of the base training set.

For **RQ3**, the main stage results do not support a consistent advantage of the selected AlexNet with GA configuration. GA achieved the best descriptive ranking on CIFAR-10 and reached 80.87% accuracy at 90% retention, only 0.08 percentage points below the 100% base training baseline. On Tiny ImageNet, however, RS obtained the best mean rank, SSRS remained competitive, and KCG reached 45.72% at only 25% retention, 0.23 percentage points below the 100% base training baseline. No subset selection method therefore dominated all experimental conditions.

The trade-offs examined in **RQ4** show that compact subsets can preserve competitive linear evaluation performance while reducing downstream training cost, but selection overhead is crucial. GA and RS wrapper searches required approximately 99–844 minutes per run, whereas SSRS was effectively instantaneous and KCG mainly incurred a reusable geometric cost. Expensive wrapper search is therefore difficult to justify for a single downstream retraining unless the selected subset is reused.

These conclusions are subject to several limitations. First, the study is restricted to linear evaluation with frozen features and does not establish that the same selected subsets would remain optimal under end-to-end fine-tuning. Second, although the experimental campaign is broad in terms of datasets, retention levels, methods, and repeated runs, this breadth should not be confused with search depth. For a base training

set of N instances and a target subset size k , the feasible search space contains $\binom{N}{k}$ subsets. Consequently, even $B = 1000$ candidate evaluations examine only a very small fraction of the available search space. At the same time, increasing the budget substantially is computationally expensive: the definitive GA and RS searches already required approximately 99–844 minutes per run. The reported rankings should therefore be interpreted as comparisons under the imposed computational budgets rather than as evidence of converged or asymptotic algorithmic superiority.

This limitation is especially important in the preliminary stage. The $B = 100$ budget is intended only for screening and is particularly restrictive for MA because population initialization and local refinement consume the same global evaluation budget. This may favor methods that use very small budgets more efficiently and may have contributed to the weak separation observed among the leading preliminary configurations. The enlarged $B = 1000$ budget in the main stage should likewise not be interpreted as a convergence guarantee. Among the 100 definitive GA/RS runs, only 10% had found their eventual best validation fitness by evaluation 100, 54% by evaluation 500, and 90% by evaluation 750. These results indicate that relevant improvements were still being found late in the available budget and that larger budgets could alter the observed relative performance of the search methods.

Additional limitations arise from the use of only five paired seeds, one final retraining per selected subset, a single validation partition, and MNIST-based preliminary screening. The small number of seeds also severely limits the resolution of pairwise statistical tests. Moreover, repeatedly evaluating up to 1000 candidate subsets against the same validation partition may induce validation set overfitting, even though the held-out test partition remains inaccessible during selection.

Future work should therefore consider additional seeds and natural image datasets, alternative frozen representations, multiple validation splits or nested evaluation protocols, and end-to-end fine-tuning. From an optimization perspective, substantially larger or adaptive search budgets, multi-fidelity evaluation, convergence-aware stopping criteria, and more efficient fitness approximations would help determine whether the relative performance of the search methods changes when deeper exploration becomes computationally feasible. Further directions include class-aware search operators, additional core set baselines, and the study of subset transfer across neural architectures.

REFERENCES

- [1] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, pp. 436–444, May 2015.
- [2] B. Sorscher, R. Geirhos, S. Shekhar, S. Ganguli, and A. S. Morcos, “Beyond neural scaling laws: Beating power law scaling via data pruning,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 19523–19536.
- [3] J. A. Olvera-López, J. A. Carrasco-Ochoa, J. F. Martínez-Trinidad, and J. Kittler, “A review of instance selection methods,” *Artificial Intelligence Review*, vol. 34, pp. 133–143, 2010.
- [4] T. Wang, J.-Y. Zhu, A. Torralba, and A. A. Efros, “Dataset distillation,” *arXiv preprint arXiv:1811.10959*, 2018.
- [5] S. García, J. R. Cano, and F. Herrera, “A memetic algorithm for evolutionary prototype selection: A scaling up approach,” *Pattern Recognition*, vol. 41, no. 8, pp. 2693–2709, Aug. 2008.
- [6] S. J. Pan and Q. Yang, “A survey on transfer learning,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.
- [7] S. Kornblith, J. Shlens, and Q. V. Le, “Do better ImageNet models transfer better?” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition*, 2019, pp. 2661–2671.
- [8] T. F. Gonzalez, “Clustering to minimize the maximum intercluster distance,” *Theoretical Computer Science*, vol. 38, pp. 293–306, 1985.
- [9] O. Sener and S. Savarese, “Active learning for convolutional neural networks: A core-set approach,” in *Proc. International Conference on Learning Representations*, 2018.
- [10] D. B. Skalak, “Prototype and feature selection by sampling and random mutation hill climbing algorithms,” in *Proc. 11th International Conference on Machine Learning*, 1994, pp. 293–301.
- [11] F. Glover and G. A. Kochenberger, Eds., *Handbook of Metaheuristics*, 1st ed. New York, NY, USA: Springer, 2003, vol. 57, International Series in Operations Research & Management Science.
- [12] P. Shaw, “Using constraint programming and local search methods to solve vehicle routing problems,” in *Principles and Practice of Constraint Programming—CP98*, M. Maher and J.-F. Puget, Eds. Berlin, Germany: Springer, 1998, vol. 1520, pp. 417–431.
- [13] D. Pisinger and S. Ropke, “Large neighborhood search,” in *Handbook of Metaheuristics*, 2nd ed., M. Gendreau and J.-Y. Potvin, Eds. Boston, MA, USA: Springer, 2010, pp. 399–420.
- [14] P. E. Hart, “The condensed nearest neighbor rule (Corresp.),” *IEEE Transactions on Information Theory*, vol. 14, no. 3, pp. 515–516, May 1968.
- [15] D. L. Wilson, “Asymptotic properties of nearest neighbor rules using edited data,” *IEEE Transactions on Systems, Man, and Cybernetics*, vol. SMC-2, no. 3, pp. 408–421, Jul. 1972.
- [16] J. Derrac, S. García, and F. Herrera, “A survey on evolutionary instance selection and generation,” *International Journal of Applied Metaheuristic Computing*, vol. 1, no. 1, pp. 60–92, 2010.
- [17] I. Triguero, D. Peralta, J. Bacardit, S. García, and F. Herrera, “MRPR: A MapReduce solution for prototype reduction in big data classification,” *Neurocomputing*, vol. 150, pp. 331–345, Feb. 2015.
- [18] B. Mirzasoleiman, J. Bilmes, and J. Leskovec, “Coresets for data-efficient training of machine learning models,” in *Proc. 37th International Conference on Machine Learning*, vol. 119, 2020, pp. 6950–6960.
- [19] K. Killamsetty, D. Sivasubramanian, G. Ramakrishnan, A. De, and R. Iyer, “GRAD-MATCH: Gradient matching based data subset selection for efficient deep model training,” in *Proc. 38th International Conference on Machine Learning*, vol. 139, 2021, pp. 5464–5474.
- [20] M. Paul, S. Ganguli, and G. K. Dziugaite, “Deep learning on a data diet: Finding important examples early in training,” in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 20596–20607.
- [21] C. Guo, B. Zhao, and Y. Bai, “DeepCore: A comprehensive library for coreset selection in deep learning,” *arXiv preprint arXiv:2204.08499*, 2022.
- [22] M. Tukan, S. Zhou, A. Maalouf, D. Rus, V. Braverman, and D. Feldman, “Provable data subset selection for efficient neural networks training,” in *Proc. 40th International Conference on Machine Learning*, vol. 202, 2023, pp. 34533–34555.
- [23] B. Zhao, K. R. Mopuri, and H. Bilen, “Dataset condensation with gradient matching,” in *Proc. 9th International Conference on Learning Representations*, 2021.
- [24] S. Lei and D. Tao, “A comprehensive survey of dataset distillation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 1, pp. 17–32, Jan. 2024.
- [25] M. Lozano, F. Herrera, N. Krasnogor, and D. Molina, “Real-coded memetic algorithms with crossover hill-climbing,” *Evolutionary Computation*, vol. 12, no. 3, pp. 273–302, 2004.
- [26] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, “Gradient-based learning applied to document recognition,” *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, Nov. 1998.
- [27] A. Krizhevsky and G. Hinton, “Learning multiple layers of features from tiny images,” University of Toronto, Toronto, Canada, Tech. Rep., 2009.
- [28] Y. Le and X. Yang, “Tiny ImageNet visual recognition challenge,” *CS231n Course Project*, Stanford University, 2015.
- [29] A. Krizhevsky, I. Sutskever, and G. E. Hinton, “ImageNet classification with deep convolutional neural networks,” in *Advances in Neural Information Processing Systems*, vol. 25, 2012, pp. 1097–1105.
- [30] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, “Aggregated residual transformations for deep neural networks,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition*, 2017, pp. 5987–5995.
- [31] M. Tan and Q. V. Le, “EfficientNetV2: Smaller models and faster training,” in *Proc. 38th International Conference on Machine Learning*, vol. 139, 2021, pp. 10096–10106.
- [32] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, “Swin Transformer: Hierarchical vision transformer using shifted windows,” in *Proc. IEEE/CVF International Conference on Computer Vision*, 2021, pp. 10012–10022.
- [33] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. 3rd International Conference on Learning Representations*, 2015.